

PIPING DESIGN

THE BASICS OF PIPING SYSTEM

A piping system is an assembly of pipe, fittings, valves, and specialty components. All piping systems are engineered to transport a fluid or gas safely and reliably from one piece of equipment to another.

Pipe is the Pipe main artery that connects the various pieces of process and utility equipment within a process plant. Although it can be considered to be the least complex component within a piping system, it is not without its peculiarities. Pipe used within a process plant designed to one of the ASME B31 codes generally is of a metallic construction, such as carbon steel, stainless steel, duplex, copper, or to a lesser degree, one of the more exotic metals like Monel or titanium.

1.1 A BRIEF HISTORY OF PIPING TECHNOLOGY

The art of design and construction of piping systems and pipelines dates back to the earliest civilizations. Its progress reflects the steady evolution of cultures around the world: the needs of developing agricultures, the growth of cities, the industrial revolution and the use of steam power, the discovery and use of oil, the improvements in steel making and welding technology, the discovery and use of plastics, the fast growth of the chemical and power industries, and the increasing need for reliable water, oil and gas pipelines.

Mesopotamia

In the valley formed by the Tigris and Euphrates (present day Iraq), between 3000 BC and 2000 BC, rose the first city-states of Ur, Uruk and Babylon. In this land, which the Greeks called Mesopotamia ("between two rivers"), man established irrigated agriculture on a grander scale than ever seen before. Networks of irrigation channels were fed by river water. At the same time, aqueducts carried potable water from springs through miles of desert. To reduce losses by evaporation, the aqueducts were partly covered or run underground. Within cities, water was distributed in cylindrical pipes made of baked clay.

China

At about the same time, and half a world away, the Chinese supplied water to their villages in bamboo pipes and used wooden plug valves to control flow. Bamboo wrapped with wax was also used to carry natural gas, while large water pipe conduits were made of hollow wood logs.

Egypt

In ancient Egypt, 3000 BC, canals were used to divert the Nile waters and irrigate fields. Drinking water was obtained directly from wells or by boiling river water. There are few reports of the use of pipes. In one instance, approximately 400 yards of copper pipes were found in the temple of Sahuri, assembled from 16" long sections made by hammering 1/16" thick sheets of copper into cylinders.

Greece

The Greeks, 1600 BC to 300 BC, used earthenware, stone, bronze and lead pipes. In many cases one end of the pipe section was tapered, while the opposite end was expanded, the tapered end of one pipe fit into the expanded end of the next section, much like today's bell and spigot joints.

Greek blacksmiths "welded" pieces of iron by hammering red-hot ends together. There is however no evidence that this type of welding was used to fabricate pipe. Whatever the fabrication technique, the pipe joints must have been reliable since the hydraulic profile of one pipeline implies that static pressure due to differences in elevation must have reached up to 300 psi at low points.

Rome

The Romans deserve special mention in the field of piping engineering. Some of their achievements in water works remained unmatched until modern times. The Roman imperial period between 400 BC and 150 AD saw the building of over 200 stone aqueducts to carry waters to three separate outlets: public baths, city fountains and a few private homes. The fountains played the role of surge tanks in case of water hammer due to sudden changes in flow. The water supply of Rome itself is reported to have been around 300 gallons per person, a high figure, even by today's standards.

Middle Ages

In Western civilization, the fall of Rome reversed the advances achieved in the science and art of piping and waterworks. Except for works by the Moors, waterworks were largely ignored in middle age Europe. Towns reverted back to wells, springs and rivers for water. As for wastewater, it was simply disposed into the streets. The exceptions appeared to have been certain abbeys that had well maintained metallic water and earthenware sewer networks. An example of color-coded flow diagram, a predecessor to modern day P&ID's (piping and instrumentation diagrams), has survived to our days. Hollowed trees were used to convey water; they were made watertight by a variety of means such as the use of sealant made of mutton fat mixed with crushed bricks.

Renaissance

Interestingly, with the invention of the printing press, one of the first books printed in the fifteenth century was Frontinus' Roman treatise on waterworks. During that period of renewal, several aqueducts were repaired and placed back in service. At the same time, metallurgy had reached a point where cast iron pipe could be produced.

The Age of Enlightenment

The waterworks of 17th and 18th century Europe are marked by advancements in pumping technology and the expanded use of cast iron pipe. Jealous of his minister's palace, the French "Sun King" Louis XIV ordered the building of 1400 fountains for his palace at Versailles. But the palace was situated on high grounds and the water had to be pumped uphill. The king entrusted the famous scientist Mariotte (1620 – 1684) to solve this problem. With a limitless budget, but on a tight schedule, Mariotte experimented with a number of pipe materials, including glass, before selecting cast iron and, in the process, perfecting the theory of strength of beams in bending. In England of the mid-18th century, the London Bridge Waterworks Company reported over 54,000 yards of wooden pipe and 1,800 yards of cast iron.

The Industrial Revolution

In the 19th century, piping technology would develop at an accelerated pace. The catalysts of this growth were the emerging oil industry, the distribution of natural gas and the increasing need for steam and water. Wood was still in use, but lap-jointed wrought iron, riveted or flanged, was taking hold. The pipe flange was perfected by S. R. Dresser in the 1880's.

Gas lighting was introduced in London in 1807, with pipelines made from musket barrels available in great numbers at the end of the Napoleonic wars. In the U.S., the first gas transmission line was installed in Baltimore in 1816.

At the threshold of the 20th century, piping technology was poised for unprecedented growth due to improvements in welding, in materials and in pumping. At the same time, standardization of materials and designs became a financial and safety necessity, and industries came to rely more on codes and standards, while national engineering societies and industry institutes became an important source of innovation and improvements.

Mechanical engineering design

Mechanical engineering design refers to the process of creating and developing mechanical systems, components, and products using engineering principles and techniques. It encompasses the entire lifecycle of a product, from conceptualization and analysis to detailed design, prototyping, testing, and manufacturing.

Mechanical engineering design involves applying scientific and mathematical principles to solve real-world problems and meet specific objectives. The primary goal is to design mechanical systems and components that are efficient, reliable, safe, and cost-effective.

The design process typically begins with identifying the problem or need that the mechanical system or product should address. This is followed by research, brainstorming, and conceptualization, where various ideas and design alternatives are explored.

Once a concept is selected, engineers use computer-aided design (CAD) software to create detailed 2D or 3D models of the mechanical components or systems. These models allow engineers to visualize and analyze the design, identify potential issues, and make necessary modifications.

During the design phase, engineers consider factors such as materials selection, structural integrity, thermal management, fluid dynamics, ergonomics, and manufacturability. They use advanced simulation and analysis tools to evaluate and optimize the design, ensuring it meets performance requirements and can withstand anticipated loads, stresses, and environmental conditions.

Prototyping and testing are critical steps in mechanical engineering design. Physical prototypes are built to verify the functionality, performance, and reliability of the design. Testing involves subjecting the prototype to various conditions and loads to identify potential flaws and areas for improvement.

Once the design is refined and validated, it moves to the manufacturing phase. Engineers work closely with manufacturing teams to ensure the design can be translated into a final product efficiently and economically. They may also provide technical specifications, instructions, and support during the production process.

Throughout the entire design process, mechanical engineers must consider factors such as safety standards, regulatory compliance, sustainability, and cost-effectiveness. They collaborate with interdisciplinary teams, including electrical engineers, materials scientists, and industrial designers, to ensure that the final product meets the desired objectives and user requirements.

In summary, mechanical engineering design involves the application of scientific and engineering principles to create mechanical systems and products. It encompasses the entire design lifecycle, from concept development and analysis to prototyping, testing, and manufacturing, with the ultimate aim of creating functional, reliable, and efficient mechanical solutions.

"Piping design" and how it relates to mechanical engineering profession

Piping design is a specialized discipline within the field of mechanical engineering that deals with the **layout, configuration, and design of piping systems**. These systems are used to transport fluids, such as liquids, gases, and slurries, from one location to another within industrial, commercial, and residential facilities.

The primary goal of piping design is to ensure the **safe, efficient, and reliable transportation of fluids while considering various factors, including process requirements, operational conditions, safety codes and regulations, maintenance accessibility, and cost considerations**. Piping systems can be found in a wide range of applications, such as oil refineries, chemical plants, power plants, HVAC systems, water supply networks, and many others.

Piping design involves several key aspects:

1. **System Layout and Routing:** Piping designers determine the most efficient and practical routes for the pipelines, considering factors such as equipment location, space constraints, and operational requirements. They must also consider the interaction between different systems and components to avoid clashes and interference.
2. **Sizing and Specification:** Piping designers calculate the appropriate pipe sizes based on factors such as flow rates, pressure drops, and allowable velocities. They select the materials for the pipes based on the properties of the transported fluids, including temperature, pressure, corrosiveness, and compatibility requirements. Pipe specifications also consider factors such as insulation, supports, expansion joints, and valves.
3. **Stress Analysis:** Piping systems are subject to various stresses, including thermal expansion, pressure loads, weight, vibration, and seismic forces. Piping designers perform stress analysis to ensure that the pipes, supports, and connections can withstand these loads without failure or excessive deformation. They use software tools and codes, such as the ASME B31.3 code for process piping, to analyze and design the system accordingly.
4. **Pipe Supports and Anchoring:** Piping designers determine the locations and types of supports needed to hold the pipes in place and prevent excessive movement or sagging. They consider factors such as pipe weight, thermal expansion, and seismic requirements. Proper support design is crucial to maintaining the integrity and stability of the piping system.
5. **Material and Component Selection:** Piping designers choose suitable materials for the pipes, fittings, valves, and other components based on factors like fluid properties, temperature, pressure, and corrosion resistance. They also consider factors such as ease of installation, maintenance, and cost-effectiveness.
6. **Documentation and Drawings:** Piping designers create detailed engineering drawings, isometric drawings, and other documentation to communicate the design intent to construction teams, fabricators, and other stakeholders. These drawings include information about pipe routing, dimensions, supports, materials, and other specifications.

Piping design is closely related to mechanical engineering as it requires a deep understanding of mechanical principles, fluid dynamics, thermodynamics, materials science, and structural analysis. Mechanical engineers with expertise in piping design play a vital role in various industries by ensuring the safe and efficient transport of fluids, optimizing energy consumption, minimizing

operational costs, and complying with regulatory requirements. They work collaboratively with other engineering disciplines, such as civil, structural, and electrical engineering, to integrate piping systems into larger projects and facilities.

The meeting point between mechanical engineering design and piping design

The meeting point between mechanical engineering design and piping design lies in the field of mechanical engineering known as **"Piping Engineering."** Piping design is a specialized discipline within mechanical engineering that focuses on the design, layout, and analysis of piping systems.

Piping engineering involves the design of piping networks that transport fluids, gases, and other substances within various industrial and commercial facilities, such as power plants, oil refineries, chemical plants, and HVAC (heating, ventilation, and air conditioning) systems.

Mechanical engineering design and piping design intersect in the following ways:

1. **Conceptual Design:** Both mechanical engineering and piping design contribute to the initial conceptualization of a system. Mechanical engineers consider the overall system requirements and the integration of various components, while piping designers focus on the specific layout and arrangement of pipes and their connections.
2. **Structural Integrity:** Mechanical engineers ensure that the overall mechanical system is structurally sound, including the supporting structures and components that interact with the piping system. Piping designers consider the forces, pressures, and temperature variations acting on the pipes to ensure their structural integrity and select appropriate materials and supports.
3. **Fluid Dynamics:** Piping design requires a thorough understanding of fluid dynamics, including flow rates, pressure drops, and fluid behavior. Mechanical engineers analyze and optimize the overall fluid flow within a system, while piping designers focus on the details of pipe sizing, routing, and pressure loss calculations to ensure efficient and reliable fluid transport.
4. **Stress Analysis:** Both mechanical engineers and piping designers perform stress analysis to assess the structural integrity of components under various operating conditions. Mechanical engineers consider the stresses on the overall system, while piping designers perform specific stress analysis on pipes, fittings, and supports to ensure they can withstand internal and external loads.
5. **Materials Selection:** Mechanical engineers and piping designers collaborate in selecting appropriate materials for the overall system and specific piping components. Factors such as fluid compatibility, corrosion resistance, and mechanical properties are considered to ensure safe and reliable operation.
6. **Codes and Standards:** Both mechanical engineering and piping design adhere to industry codes, standards, and regulations. Mechanical engineers ensure that the overall system complies with relevant codes, while piping designers focus on meeting specific piping codes and standards.

Overall, mechanical engineering design and piping design work together to create integrated and efficient mechanical systems that incorporate safe and reliable piping networks. Collaboration and communication between mechanical engineers and piping designers are crucial to ensure a successful and optimized design.

What are the specific functions of the mechanical engineer with regards to building piping design

As a mechanical engineer involved in building piping design, you would have several specific functions. Here are some of the key responsibilities and tasks associated with this role:

1. **System Design:** Mechanical engineers play a crucial role in designing the piping system for buildings. This includes determining the layout, sizing, and routing of pipes to meet the building's requirements, such as water supply, heating, cooling, and drainage systems. They consider factors like pressure, flow rates, material compatibility, and safety codes while designing the system.
2. **Material Selection:** Mechanical engineers evaluate and select appropriate materials for piping systems based on the specific application, fluid characteristics, temperature, and pressure conditions. They consider factors such as corrosion resistance, durability, cost, and ease of installation when choosing materials like copper, steel, PVC, or PEX.
3. **Calculations and Analysis:** Engineers perform calculations to determine pipe sizing, pressure drop, and flow characteristics within the system. They use engineering principles and formulas to analyze the system's behavior under different operating conditions and ensure it meets performance and safety standards.
4. **Code Compliance:** Mechanical engineers ensure that the piping design adheres to relevant building codes, standards, and regulations. They stay up-to-date with local, national, and international codes to ensure compliance and maintain the safety and integrity of the piping system.
5. **Collaboration with Other Disciplines:** Mechanical engineers work closely with architects, civil engineers, structural engineers, and other professionals involved in the building design process. They coordinate their designs with other disciplines to ensure proper integration of the piping system within the overall building design.
6. **Equipment Specification:** Mechanical engineers specify the necessary equipment and components for the piping system, such as pumps, valves, fittings, and supports. They consider factors like capacity, pressure ratings, efficiency, and compatibility with the overall system design.
7. **Documentation and Drawings:** Engineers create detailed drawings, schematics, and specifications for the piping system. These documents include dimensions, material specifications, pipe routing, and support requirements. Accurate documentation is vital for construction, installation, and future maintenance of the system.
8. **Construction Support:** Mechanical engineers may provide support during the construction phase, working closely with contractors and construction teams. They may review installation plans, conduct site inspections, and ensure that the piping system is installed correctly according to the design specifications.
9. **System Testing and Commissioning:** Engineers participate in testing and commissioning activities stem. They may conduct pressure tests, flow tests, and functional checks to ensure proper operation and identify any issues that need to be resolved.

Overall, the mechanical engineer's role in building piping design encompasses a wide range of responsibilities, from system design and material selection to calculations, compliance, collaboration, documentation, and construction support.